

# RUTURAJ MARATHE

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## SUMMARY

Backend Software Engineer with 3+ years of experience owning backend systems end-to-end, from API design and implementation to infrastructure setup and deployment as part of Architecture and Innovation Team.

## WORK EXPERIENCE

### Visteon Corporation

Van Buren, MI, USA

Software Developer – AI & Innovation, IT, Automation

Jun 2025 – Present

- Designed and implemented a **microservice-based** system to automatically generate complex automotive test reports, reducing effort by **5000 hours per operator annually**.
- Integrated **OCR and multimodal models** to extract structured data from block diagrams and generate Bills of Materials, improving process efficiency by 70%.
- Built an automated pipeline to **cost the BOM files**, reducing hardware architecture turnaround time.
- Refactored and **optimized production SQL queries** by improving joins, applying push down filters and cutting data fetch latency from 20 seconds to 5 seconds.

### Intel Corporation

Santa Clara, CA, USA

Software Engineer Intern – Machine Learning, Automation, Python

May 2024 – Dec 2024

- Automated the **defect creation programmatically** to streamline bug filing, saving 70% of hardware engineers time by populating the details via **REST API**.
- Delivered a Machine Learning pipeline that runs **DBSCAN clustering** to group SoC design paths, auto generate waiver codes, leading to a left-shift in fixing Setup/Hold violations.
- Developed AI solution by training and inferencing an **XGBoost model** to predict delays, reducing IO budgeting iterations from 10 to 6.
- Presented **POC AI use-cases** in Electronic Design Automation to boost the productivity of SoC designers.

### Visteon Technical and Services Center

Bangalore, KA, India

Software Engineer – Embedded Linux, C, Java, SQLite, IVI

Jan 2021 – Jul 2023

- Designed and deployed a media file favorites feature across **Android middleware**, enabling easy access to multimedia files for 500+ **Harley-Davidson** IVI users.
- Developed **AIDL interfaces** and business logic for 3 multimedia features enabling efficient **Inter Process Communication** between android services in a complex multi-threaded stack.
- Implemented Factory Reset to enforce **disaster recovery** by re-syncing media files enforcing 100% reliability.
- Resolved 45% of **JIRA defects** in audio playback & browsing by root cause analysis of logs and IVI testing.

## SKILLS

|                                     |                                                                                                           |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Backend Engineering:</b>         | Python, C/C++, FastAPI, Microservices, Automation Pipelines, SQL Query Optimization, Data Processing, ETL |
| <b>Cloud &amp; Infrastructure:</b>  | Linux, AWS (EC2, S3), Azure, Docker, CI/CD, Jenkins, Git, GitLab                                          |
| <b>Embedded &amp; Platforms:</b>    | Embedded Linux, Android Middleware, AIDL, IVI Systems, IPC Mechanisms                                     |
| <b>AI / Machine Learning:</b>       | OCR, NLP, XGBoost, DBSCAN, GenAI, Streamlit, Scikit-learn                                                 |
| <b>Databases:</b>                   | SQL, Oracle, SQLite, Data Modeling, Query Performance Tuning                                              |
| <b>Awards &amp; Certifications:</b> | Visteon Excellent Performance Award, AWS Cloud Practitioner                                               |

## KEY PROJECTS

### DoctorAssist.AI (Streamlit, GenAI, NLP, Gemini API)

Apr 2025

- Engineered a natural language-to-SQL query engine to streamline healthcare data access, enabling clinicians to retrieve patient information efficiently using conversational queries, improved access speed by 40%.

### Used Car Data Analysis (AWS, EC2, S3, boto3, PySpark)

Nov 2024

- Architected a distributed data pipeline to analyze large-scale used car listings and extract pricing insights, improving dealership pricing strategy efficiency by 25%.

## EDUCATION

**Master of Science in Computer Science** - Software Engg, Applied Machine Learning

GPA 3.87

Indiana University Bloomington, IN, United States

May 2025

**Bachelor of Engineering in Computer Science** - Algorithms, OS, Networks

GPA 8.08

Visvesvaraya Technological University, KA, India

August 2020